

Ex No: 2b

IMPLEMENTATION OF UNIFICATION AND RESOLUTION
ALGORITHM

Aim:

To implement unification and resolution algorithm using python.

Procedure:

Define unify, unify_var, and occurs_check functions to recursively unify terms and variables while updating substitutions and checking for cyclic dependencies.

Define the negate helper function to invert literals for resolution processing

Define the resolve function to identify complementary literals between two clauses and produce new resolvent clauses.

Implement the resolution function by combining the knowledge base with the negated query, iteratively generating resolvents until an empty clause is found or no new clauses are produced

Construct a sample knowledge base and query, execute the resolution function, and output whether the query is satisfiable.

Program:

Unification and Resolution in Python

```
def unify(var1, var2, subst):
```

```
    """Unify two variables or terms with substitution."""
```

```
    if subst is None:
```

```
        return None
```

```
    elif var1 == var2:
```

```
        return subst
```

```
    elif isinstance(var1, str) and var1.islower():
```

```
        return unify_var(var1, var2, subst)
```

```
    elif isinstance(var2, str) and var2.islower():
```

```
        return unify_var(var2, var1, subst)
```

```
    elif isinstance(var1, list) and isinstance(var2, list) and len(var1) == len(var2):
```

```
        return unify(var1[1:], var2[1:], unify(var1[0], var2[0], subst))
```

```
    else:
```

```
        return None
```

```

def unify_var(var, x, subst):
    """Handle unification for variables."""
    if var in subst:
        return unify(subst[var], x, subst)
    elif x in subst:
        return unify(var, subst[x], subst)
    elif occurs_check(var, x, subst):
        return None
    else:
        subst[var] = x
        return subst

def occurs_check(var, x, subst):
    """Check for cyclic dependencies in substitution."""
    if var == x:
        return True
    elif isinstance(x, list):
        return any(occurs_check(var, xi, subst) for xi in x)
    elif x in subst:
        return occurs_check(var, subst[x], subst)
    return False

def resolution(kb, query):
    """Apply resolution algorithm to check satisfiability."""
    clauses = kb + [negate(query)]
    while True:
        new_clauses = []
        for i, ci in enumerate(clauses):
            for j, cj in enumerate(clauses):
                if i >= j:

```

```

        continue
    resolvents = resolve(ci, cj)
    if [] in resolvents:
        return True # Query is satisfiable
    for res in resolvents:
        if res not in new_clauses:
            new_clauses.append(res)
    if all(new in clauses for new in new_clauses):
        return False # Query is not satisfiable
    clauses.extend(new_clauses)

```

```

def negate(literal):
    """Negate a literal or a list of literals."""
    if isinstance(literal, list):
        return [negate(lit) for lit in literal]
    if literal.startswith("~"):
        return literal[1:]
    else:
        return f"~{literal}"

```

```

def resolve(clause1, clause2):
    """Resolve two clauses to find resolvents."""
    resolvents = []
    for lit1 in clause1:
        for lit2 in clause2:
            if lit1 == negate(lit2):
                new_clause = list(set(clause1 + clause2))
                new_clause.remove(lit1)
                new_clause.remove(lit2)
                if new_clause not in resolvents:
                    resolvents.append(new_clause)

```

```

    return resolvents
if __name__ == '__main__':
    # Example knowledge base and query
    knowledge_base = [
        ["~P", "Q"],
        ["P"],
        ["~Q", "R"],
        ["~R"]
    ]
    query = ["R"] # Representing the query as a list
    print("Knowledge Base:", knowledge_base)
    print("Query:", query)
    result = resolution(knowledge_base, query)
    if result:
        print("The query is satisfiable.")
    else:
        print("The query is not satisfiable.")

```

Output:

Knowledge Base:

```

[
    ["~P", "Q"],
    ["P"],
    ["~Q", "R"],
    ["~R"]
]

```

Query:

```

["R"]

```

Knowledge Base: [['~P', 'Q'], ['P'], ['~Q', 'R'], ['~R']]

Query: ['R']

The query is not satisfiable.

```
Python 3.11.4 (tags/v3.11.4:8e8a0a4, Jun 6 2023, 13:03:39) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> ===== RESTART: C:/Users/Param/Desktop/resolution.py =====
      The query is not satisfiable.
>>> |
```